

Sean Luka Girgis

Senior Data Engineer | Python, PySpark, Spark SQL, AWS & Data Governance

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Professional Summary

Former Citi Senior Data Engineer with 20+ years of enterprise technology experience and strong hands-on background in Python, SQL, PySpark/Spark, ETL/ELT pipelines, AWS data platforms, Redshift, Oracle, data quality, governance-aware delivery, reporting, performance tuning, and production troubleshooting. Experienced building scalable telemetry and analytics pipelines, optimizing SQL and pipeline performance, documenting metadata and metric definitions, and supporting regulated enterprise reporting systems. Best aligned to Citi data engineering roles requiring Python, PySpark, Spark SQL, ETL/ELT, AWS/Redshift, data quality, governance, analytics pipelines, and stakeholder collaboration; Databricks, Delta Lake, Snowflake, Starburst/Trino, Apache Iceberg, and Ab Initio are positioned as adjacent or active-learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, analytics, and downstream reporting workloads.
- Developed Spark/PySpark and SQL-based transformation logic for large-scale telemetry processing, historical retention, schema-aware reporting, and performance-oriented analytics datasets.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Built data quality, reconciliation, and validation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve consistency, completeness, and trust in pipeline outputs.
- Supported governance-aware delivery by documenting metric definitions, data assumptions, lineage-style notes, operational runbooks, access considerations, and stakeholder-facing reporting logic.
- Partnered with infrastructure, application, analytics, architecture, and business stakeholders to gather requirements, troubleshoot data issues, and deliver production-ready data products aligned to Citi standards.
- Prepared model-ready and analytics-ready datasets supporting Prophet/scikit-learn forecasting workflows, capacity insights, and AI/ML-adjacent experimentation in a regulated banking environment.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.
- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and data/reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, operational analytics, dashboards, and production reliability.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze data, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, and reporting access.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine

Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.

Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Validation

AI-Powered Job Search Pipeline

Built an AI-assisted Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking using Claude Code, Grok/xAI APIs, FAISS, and sentence-transformers.

Technologies: Python, Claude Code, Grok / xAI API, FAISS, Sentence Transformers, JSON, Quality Gates